CS2090b · Data Science 2 | Milestone 2

Hallucination in Legal RAG Chatbots

Data Wrangling & Project Redefinition

April 10, 2026

Alex Oort Alonso · Allan Korir · Phong Le · Brit Biddle

Harvard University | COMPSCI 1090B

Motivation — Why This Problem Matters

Mata v. Avianca Airlines (2023)

An attorney submitted a legal brief to a federal court citing six case precedents — all generated by ChatGPT. None of the cases existed. The court sanctioned the attorney and dismissed the citations. This is what hallucination looks like in the real world.

Legal Liability

Attorneys face sanctions, malpractice claims, and loss of bar standing for filing AI-hallucinated citations.

Client Harm

Clients lose cases based on fabricated law. In high-stakes contexts — criminal, immigration, civil rights — errors are devastating.

Trust Collapse

One hallucination incident erodes trust in all legal AI tools, slowing beneficial adoption across the profession.

Our question: Which retrieval architecture best minimizes contradictions and evidence gaps in AI-generated legal responses?

Project Overview — Retrieval Architectures & Evaluation

BM25	BGE-M3	Hybrid	Legal-BERT
Lexical Baseline	*Dense Retrieval*	*Primary Candidate*	*Reference Model*
Sparse term-frequency retrieval (non-ML)	BAAI/bge-m3 CLS pooling (bfloat16)	BM25 + BGE-M3 + CrossEncoder rerank (e.g. bge-reranker)	Domain-specific bi-encoder (512 sub-words)

Generator held constant across all architectures: API-based LLM (gpt-5.4-nano)

3 Hallucination Measurements

Tier A	**Retrieval Grounding**	Hit@k, MRR, and NDCG@10 via LePaRD expert-annotated citation pairs (10K–50K queries)
Tier B	**NLI Contradiction Detection**	DeBERTa-v3-large classifies each atomic claim — contradiction rate per claim + per 1K tokens (1K stratified queries)
Tier C	**Hard Citation Hallucination**	Every citation is looked up in a local SQLite index built from the CourtListener corpus. If NULL ⟶ Hard Citation Hallucination logged (NLI skipped).

Data Sources

62 GB

Raw CSV Download
(4 CourtListener files)

1,456,611 opinions

Federal Appellate
Opinions Verified in 159 shards

4 M

LePaRD Training
Pairs (HuggingFace)

CourtListener — Federal Appellate Corpus

Primary training corpus · License: CC BY-ND 4.0

opinions-2026-03-31.csv.bz2

dockets-2026-03-31.csv.bz2

opinion-clusters-2026-03-31.csv.bz2

courts-2026-03-31.csv.bz2

After filtering to federal appellate courts: 1,456,611 opinions in 159 shards

LePaRD — Legal Passage Retrieval Dataset

Retrieval + hallucination evaluation · Source: HuggingFace

- **Total pairs:** ~4 million U.S. federal judge citations
- **Training cap:** Revised Scope: cap raised to 4M train + 1M held-out test
- **Eval queries:** 10K–50K retrieval · 1K generation
- **Pinned to:** 40-char SHA revision (mutable refs rejected)
- **Pairs type:** Ground-truth passage-level query–document
- **Compatibility:** Cross-checked vs. CourtListener corpus

Data Acquisition & Wrangling Pipeline — 9-Cell Colab Notebook

Platform: Google Colab Pro · GPU: NVIDIA A100-SXM4-80GB (85.1 GB VRAM) · Runtime: Python 3.11.9 · Drive persistence: all artifacts symlinked to Google Drive

C2

Bootstrap

Clone repo · install uv · create pinned Python 3.11.9 venv · verify deps

C3

Preflight Gate

20+ contract tests · CUDA verified · 85.1 GB VRAM confirmed · seeds pinned

C4

Download CSVs

4 CourtListener files from S3 · 62 GB total · Drive fast-path on reconnect

C5

Filter + Extract

courts → dockets → clusters → opinions chain · 1,465,484 opinions → 159 shards · SHA-256 manifest

C6

Corpus Probe

8-gate RAG readiness · Polars full-scan · probe v2.5.11 · 303 contract tests

C7

LePaRD Ingest

HuggingFace download · 40-char SHA pinned · atomic write + .sha256 sidecar

C8

Compatibility Audit

LePaRD × CourtListener cross-check · usable gold pair rate reported · CI-gate capable

C9

Quality Gate

48-core parallel NaN/encoding/parse audit across all 159 shards · verdict: CLEAN

EDA & Corpus Statistics

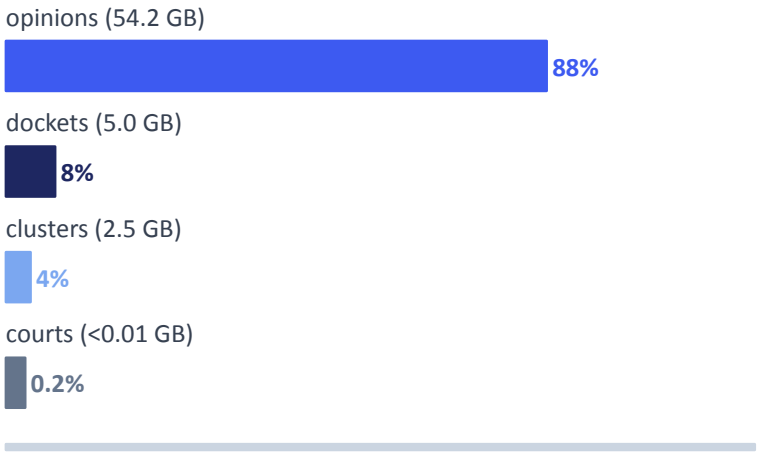
159
JSONL Shards
(SHA-256 each)

7.6 GB
Processed
Corpus Size

~50K
Avg. Characters
per Opinion

CLEAN
Data Quality
Audit Verdict

Raw Download Size by File



Total: 61.6 GB → filtered to 7.6 GB (87% reduction)

Key EDA Findings

- Text length is right-skewed — most opinions 5K–20K chars; long tail of full opinions >50K chars
- 9th Circuit is ~22% of corpus — largest single circuit; significant class imbalance across circuits
- ~6% of records had no extractable text (sealed opinions, docket-only entries, empty HTML fields)
- Pre-2000 opinions show high OCR noise — low entropy, garbled Unicode, broken sentence structure
- ~14% of opinions have zero citations — advisory-only flag; not excluded from retrieval corpus
- LePaRD compatibility audit confirmed usable gold pair rate across CourtListener federal appellate subset

8-Gate RAG Readiness Probe · All Gates: PASSED

src.dataset_probe v2.5.11 · 303 contract tests frozen in ProbeConfig · Full Polars memory-mapped scan · ~30 seconds runtime

Gate	Name & Description	Threshold
Schema	Schema Compliance — 17 STAGE3_REQUIRED_FIELDS enforced via typed Pydantic contracts · lazy loading · hard gate	0 violations
A7	Text Source Breakdown — Priority chain resolves text field (plain_text → html_with_citations → html). ≥80% from known formats	≥80% known
A8	Text Length Distribution — ≤25% docs below 1,500 char RAG threshold. Short records soft-excluded, not deleted.	≤25% short
A9	Citation Count (Advisory) — Flags zero-citation documents. ≤20% allowed. Advisory only — no blocking on failure.	≤20% zero-cit.
A11	Tokenizer Chunk Count — BGE-M3: 1,024 sub-word chunks / 128 overlap / 896 stride. Median ≥2 chunks. Sample n=200.	median ≥2
A12	Citation Anchor Survival — Regex on normalized text (50K char cap). ≥60% docs must retain detectable citation patterns. n=500.	≥60% survive
A13	Sentence Density — spaCy segmentation. ≤15% docs below 20-sentence minimum. Catches OCR/boilerplate stubs. n=200.	≤15% sparse
B6	Text Entropy Distribution — Character-level Shannon entropy. ±1.0 bit tolerance from corpus mean. Flags boilerplate and OCR noise.	±1.0 bit

Data Challenges & Wrangling Decisions

① Missing / Null Text Fields (~6%)

Problem: Sealed opinions, docket-only records, and empty HTML fields — no readable text at all.

Fix: Row validator enforces field presence chain. Invalid rows skipped via iter_valid_rows() — skipped count logged to W&B but records not deleted.

② OCR Noise (pre-2000 Opinions)

Problem: PDF-scanned opinions introduce broken Unicode, character substitutions, garbled symbols, and very low entropy scores that pollute embeddings.

Fix: Gate B6 entropy filter flags worst offenders. repair_text_length.py corrects truncation artifacts. Soft-exclude with warnings — not hard deleted.

③ CourtListener and LePard Matching

Problem: Only 27% of the citations found in LePard (the labelled dataset we're using) have exact text matches to the opinions in CourtListener, which means that the majority of citations could not actually be retrieved by the RAG system.

Fix: Filter LePard queries to only include those with a valid citation within the CourtListener dataset which the retrieval system can access. Could also just evaluate on citing a general case/opinion as opposed to a specific paragraph.

④ Scale & Throughput (1.46M opinions)

Problem: BGE-M3 tokenization and chunking across 1.46M documents is CPU-intensive. Naive loops time out. GPU contention with embedding phase.

Fix: Polars mandatory for full-corpus scans (CPU-only, zero GPU contention). Reservoir sampling for expensive gates (n=200–500). DVC shards for incremental reruns.

Project Redefinition — 4 Changes from Original Proposal

Based on data wrangling findings and feasibility constraints, we made four deliberate scope changes from the original project proposal:

Data Source

Before: Harvard CAP (6.7M decisions) + full Pile of Law as primary corpora

After: CourtListener federal appellate subset (~1.46M opinions) — scoped to ~500K for M2 wrangling

Why:

CourtListener provides structured JSONL access + SHA-256 manifests. CAP full corpus exceeds storage and GPU budget.

Architectures

Before: 5 methods: TF-IDF/Word2Vec, CNN encoder, RNN/LSTM, BERT bi-encoder,

After: 3 core + 1 reference: BM25, BGE-M3, Hybrid (+ CrossEncoder reranker), Legal-BERT reference

Why:

CNN/BiLSTM are outdated baselines; KG retrieval deferred. Modern dense retrievers give stronger, more meaningful comparisons.

Hallucination Eval

Before: Human-annotated hallucination rate, faithfulness score, F1 on generated answers

After: Automated: DeBERTa-v3-large NLI (Tier B) + SQLite citation lookup (Tier C) — no human annotation

Why:

Human annotation at 1.46M-document scale is not feasible. DeBERTa achieves near-human NLI agreement; SQLite removes API bottleneck.

Retrieval Metrics

Before: Recall@k and Precision@k

After: Hit Rate@k + MRR

Why:

Hit Rate @ k reflect the structure of the testing set (LePard), which only has 1 citation per query, and MRR provides additional information regarding retrieval accuracy.

How We Measure Hallucination — 3 Measurements

Each retrieval architecture is evaluated on 4 complementary hallucination signals across 3 tiers:

Tier A

Retrieval Precision — Hit Rate@k · MRR

How: LePaRD expert-annotated citation pairs provide ground truth. Each architecture retrieves top-k passages for 10K–50K queries. Recall@k measures whether the gold passage is in the results; MRR measures how highly it's ranked; NDCG@10 weights rank position.

Why it matters:

If the retriever can't find the right passage, the generator will hallucinate regardless. This is the upstream guard.

Tier B

NLI Contradiction Rate — per claim & per 1K tokens

How: 1,000 stratified queries (by circuit) → generate answer → extract atomic claims → DeBERTa-v3-large-mnli classifies each claim against retrieved chunks (512-token windows, 64-token stride). Contradiction normalized 2 ways. Zero-claim responses tracked separately.

Why it matters:

Measures whether the generated text directly contradicts the source passages — the purest hallucination signal.

Tier C

Hard Citation Hallucination — citation resolves to NULL

How: Every citation in a generated answer is extracted and looked up in a local SQLite index built from the CourtListener corpus. If the cited case ID resolves to NULL — the case simply does not exist — it's a confirmed hallucination. No API calls; read-only, check_same_thread=False.

Why it matters:

Directly replicates the Mata v. Avianca failure mode — fabricated case citations.

✓ **Data Pipeline Complete — 50% of Project Effort Done. Modeling & Evaluation Remaining.**

Milestone 3 — What Comes Next

01

Index & Embed

Build BM25 index + BGE-M3 FAISS index over 1.46M opinions (1024 sub-word chunks)

02

Run All 4 Architectures

BM25 · BGE-M3 · Hybrid + CrossEncoder · Legal-BERT — same pre-chunked payloads

03

Tier A — Retrieval Eval

Recall@k · MRR · NDCG@10 on 10K–50K LePaRD queries → head-to-head ranking

04

Tier B — Hallucination Detection

1K stratified queries → DeBERTa NLI → contradiction rate per claim + per 1K tokens

05

Tier C — Citation Verification

SQLite lookup → hard hallucination (NULL) (CitationFound_NoLocalSupport)

06

Demo-Ready Chatbot

Full pipeline producing grounded legal answers — retrieval + generation + hallucination verdict